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UNIVERSITI
TEKNOLOGI
PETRONAS

FINAL YEAR PROJECT GUIDELINES

FOR

SUPERVISORS AND STUDENTS

UNIVERSITI TEKNOLOGI PETRONAS
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1. INTRODUCTION

All final-year students at UTP are required to undertake a Final Year Project (FYP) course, which is a research / project-based subject. This course is a compulsory component of all programs offered at UTP.

This course requires students to do research, project and/or development work in their respective discipline, especially on real-world problems which would motivate them to produce practical solutions. It is an opportunity for students to use the tools and techniques of problem-solving to solve the problems they have encountered. This hands-on approach allows students to learn through direct, practical experience. Management concepts which provide students with skills required for managing a project are also incorporated. Thus, the students are expected to be well rounded by mastering various useful disciplines, which will enable them to participate and prepare for future employment.

Working under the guidance of a Supervisor / Supervisors, students may shape the direction of their goals, as well as gain better understanding of the responsibilities involved in conducting research or projects. The undertaken research / project can also be used as a basis for job employment by fully exploiting the learning process they have gone through, the skills they have gathered and the experience they have gained from the research / project.

This guideline is prepared for students and Supervisors to enable them to execute their respective roles and responsibilities in an effective manner, hence benefiting both parties, in successfully achieving the course learning outcome for the subject. The mapping of Program Outcome / Learning Cluster towards the Course Learning Outcome is given in **Appendix 1.1**.

2. PURPOSE

The purpose of the project is to develop a framework, which will enhance students' skills in the process of applying knowledge, expanding thoughts, solving problems independently and presenting findings through minimum guidance and supervision. The requirement for FYP students to write and submit a dissertation and a technical paper is vital for several reasons, including career preparation, academic success, and personal development. It also plays a vital role in UTP's educational process by

guaranteeing that the academic program is of the highest calibre and meets international standards.

3. SCOPE OF WORK

The project can be in the form of laboratory experiments, computer programming, modelling, simulations, analysis and product design. The area and scope of the project should be narrowed down so that the project is feasible and could be completed within the allocated time frame. The project work must exhibit an element of originality and indicates the maturity level for final year bachelor's degree programme.

4. RESPONSIBILITIES

A team of FYP Committee comprising of the Chairperson, FYP Coordinator, Cluster Leaders, Supervisor, Co-Supervisor and Panel of Examiners is formed to manage the Final Year Project (**Appendix 1-2)**. They should work closely with each other in managing and monitoring students' work progression. It is the responsibility of each party to ensure that the project is completed and delivered within the project time frame. The team responsibilities are as indicated in the *Final Year Project Process Flow* in **Appendix 1-3**.

4.1 FYP Committee

The Chairperson of the FYP Committee for each Programme is the respective Department Chair. The Chairperson of the FYP Committee is responsible in appointing the FYP Coordinator and the Committee Members.

4.1.1 The main tasks and responsibilities of the **FYP Coordinator** are as follows:

(a) To execute the milestone for the managing the final year project as shown in **Appendix 2-1** and **2-2**.

(b) To identify the students who have registered for the final year project.

(c) To distribute the *FYP Guidelines* to students and Supervisors.

(d) To plan and manage the final year project process.

(e) To collect the suggestions of project title proposals from the lecturers, other staff and students. Please refer to **Form 01A or Form 01B** in **Appendix 3-1**. (Appendix 3-1a or Appendix 3-1b).

(f) To submit all project proposals to the FYP Committee for approval and to assign Supervisor(s).

(g) To release the list of approved projects for students' selection.

(h) To collect the list of project titles selected by students. Please refer to **Form 02** in **Appendix 3-2**.

(i) To forward **Form 02** to the FYP Committee for approval.

(j) To release the FYP Committee's decision on project titles and Supervisor(s).

(k) To brief the students and Supervisors on the project requirements based on the guidelines.

(l) To co-ordinate a committee meeting to appoint a panel of Examiners.

(m) To endorse **Form 03** and forward the form to the FYP Committee Chairperson for approval to purchase consumables and equipment, technical support, nomination of advisors and other related resources for the project. Please refer to **Form 03** in **Appendix 3-3**.

(n) To arrange students' oral presentation prior to Week 14 of each semester.

(o) To collect the interim report/dissertation final drafts from students and forward it to the Examiners for evaluation during oral presentation during the semester.

(p) To compile the students' assessment marks.

(q) To collect two (2) hard cover copies and softcopies of final dissertation from students.

(r) To compile and retain **Forms 01**, **02**, **03**, **04**, **05**, **06**, **07**, **08**, **09**, **10**, **11, 12, 13** and **14** for at least one (1) year after graduation for auditing purposes.

4.1.2 The tasks of the **FYP Committee Chairperson** and **Committee Members** are as follows:

(a) The Chairperson is responsible for all final decisions based on the committee's recommendations.

(b) To ensure the initial project proposal is relevant to student's field of study.

(c) To approve project title proposals.

(d) To assign project main Supervisor and Co-Supervisor (if any).

(e) To assign project titles to students.

(f) To appoint panel of Examiners.

(g) To endorse final project grades.

(h) To approve proposal to purchase consumables and equipment, technical support, nomination of advisors and other resources related to projects.

4.2 Supervisor

Students will be supervised by one main Supervisor, who is knowledgeable in the relevant field of expertise. Their responsibilities can be summarised as follows:

(a) To work together with students until the submission of the final report.

(b) To assist students on the accessibility of the tools needed for the project.

(c) To monitor the schedule and progress of the students and their projects.

(d) To assist and guide students on the project and the preparation of an interim report and a dissertation according to the approved format.

(e) To assess students' performance, including conducting continuous assessments. Please refer to **Form 04, 06, 07, 09, 10 and 11** in the relevant Appendices.

(f) To deliver their part of evaluation. Please refer to grading structure of Table 2 in Section 7.

(g) To nominate a Co-Supervisor, if required, and notify the FYP Coordinator.

(h) To forward a proposal to the FYP Coordinator for endorsement on purchasing of consumables and equipment, technical support, nomination of advisors and other related resources for the project. Please refer to **Form 03** in **Appendix 3-3**.

(i) To nominate Examiners to FYP Committee.

(j) To compile and retain all evaluation for at least one (1) year after graduation for auditing purposes.

4.3 Co-Supervisor

A Co-Supervisor may be nominated by the Supervisor to assist students in very specialized areas. The Co-Supervisor can be internal or external (from outside UTP). The external Co-Supervisor must be officially appointed by the FYP Committee. The Co-Supervisor's responsibility is to guide students in solving specific tasks within his/her expertise as and when necessary, within the project time frame. In certain cases, the Co-Supervisor may represent the Supervisor in oral presentations and evaluate the students' progress and final dissertation.

4.4 Panel of Examiners

The FYP Committee appoints the Panel of Examiners. The Examiners will function as project evaluators who are responsible to evaluate the oral presentation and dissertation. Refer to **Table 2** in **Section 7**. Examiners for FYP1 and FYP2 will be selected internally, with the addition of industry Examiners who will also be invited to assess FYP2 students. If it is not possible to get an External Examiner from the industry, the FYP Committee can appoint another lecturer from the same programme as a replacement.

4.5 Students

In completing the project, students are required to demonstrate their ability to integrate fundamental knowledge in developing techniques, methods and analyses. Students can propose their own project title to be vetted by the FYP Committee members or select a selection of titles endorsed by the Committee members. Students are also required to work independently through exercising self-discipline, self-management and job co-ordination while undertaking the project. If working in a group, the students are expected to exercise teamwork, co-operation, and trustworthiness to ensure the success of the project.

It is compulsory for students to complete **ALL** assessment processes as follows:

(a) Proposal Defence
(b) FYP 1 Interim Report
(c) Dissertation
(d) Technical Paper
(e) Project Presentation (product demo, where applicable)

A student will be given a grade of F for failing to do so.

Students' responsibilities include the following:

(a) To select a project topic on their own, or the one suggested by the Supervisor/s.

(b) To produce a preliminary report which clearly defines the objective(s) and the scope of project work.

(c) To plan meeting schedules with their Supervisor/s.

(d) To adhere to the meeting schedules with the Supervisor/s for the purpose of updating their progress and seeking advice on project matters.

(e) To be responsible in finding alternative solutions for problems encountered such as computer crashes and instrument failure.

(f) To submit all reports on time with no exception.

5. PROJECT PROCESS FLOW

The *Final Year Project Flow Sheet* is shown in **Appendix 1-3**.

5.1 Submission of Titles and Project Synopsis

(a) Lecturers may submit upon invitation by the FYP Coordinator their project titles and proposals according to their interest and expertise to the FYP Coordinator.

(b) Students are also encouraged to propose their own project proposals based on their interest and experience.

(c) All proposals must be submitted to the FYP Coordinator for each department using **Form 01** as in **Appendix 3-1** before the start of the new semester.

5.2 Approval on Project Proposal and Assignment of Supervisor/s

(a) The FYP Coordinator will forward the list of proposed project titles to the FYP committee for vetting and endorsement.

(b) The project proposal will be reviewed by the FYP Committee to ensure the viability of the project.

(c) The proposer will be called by the FYP Committee, if any clarification is required.

(d) Once proposed projects and titles are approved, the FYP Committee will endorse the projects to be released to the students.

(e) The FYP Coordinator will advertise the list of approved projects and titles to the students.

5.3 Selection of Project Titles

(a) Students who do not submit a project proposal or have their proposals rejected by the FYP committee of the department, will have to select **a maximum of three (3) titles** from the given list in any order of preference and submit it to the FYP Coordinator using **Form 02** in **Appendix 3-2**.

(b) The FYP Coordinator will forward the students' project title selections to the FYP Committee members, who will then assign approved projects or Supervisors to the students.

5.4 Allocation of Approved Project Titles / Assigned Supervisors

Students will be informed by the FYP Coordinator on the awarded project or assigned Supervisors.

5.5 Approval for Purchase and Usage of Resources and Services

(a) If required, the Supervisor will forward **Form 03** (**Appendix 3-3**) to the FYP Coordinator for the purchase of consumables and equipment, nomination of Advisor(s), request for technical support, a visit and other resources related to the project as necessary. The FYP Coordinator or Laboratory Executive will then forward the form to the FYP Chairperson for approval.

(b) For projects involving experimental works, students are required to register with the Laboratory Management Department (LMD). Students are expected to conduct risk assessment, comply with laboratory rules and regulations, and perform good laboratory practices.

5.6 Progress Assessment (FYP I and FYP 2)

Progress assessment is a **continuous assessment** carried out by Supervisors for student undertaking FYP 1 and FYP 2. It is used to assess the attainment of specific course learning outcome, as well as helping Supervisors to monitor the research / project progress of students. Student and Supervisor need to agree on the research / project activities at the start of the research /project work. It is

recommended for students to meet their Supervisors regularly (at least once every two weeks) in ensuring good research / project work progress.

5.7 Proposal Defence (FYP 1)

In this seminar, students should be able to verbally report the progress of their project to the Supervisor, fellow students and other lecturers attending the seminar. The seminar is also an avenue for the student to get feedback on how to improve their project. The seminar can be organised in small or large groups. The students need to do the oral presentation to the Supervisor and internal Examiner as appointed by the FYP Committee. The Examiners will evaluate the presentation and submit the marks to the FYP Coordinator using **Form 05 (Appendix 3-5**). If the students fail to attend the oral presentation, the students will be barred and will not be able to continue the course.

5.8 Submission of Interim Report (FYP 1)

Towards the submission, students should be able to:

(a) Write an abstract of the study

(b) Identify the problem statement, objective and scope of the study

(c) Write the literature review

(d) Explain the methodology to be used in the study

(e) Plan the deliverables for the whole project (FYP 1 and FYP 2)

The procedures for submission are as follows:

(a) Students will submit an *Interim Report* to the FYP Coordinator for endorsement prior to submission to the Supervisor and internal Examiner.

(b) The Supervisor and internal Examiner will evaluate the *Interim Report* and submit the marks to the FYP Coordinator using **Form 08** as in **Appendix 3-8**.

5.9 Submission of Technical Paper (FYP 2)

Students are required to submit a technical paper along with the Dissertation. The technical paper should consist of an Abstract, Keywords, Introduction, Literature Review, Methodology, Results, Discussion, Conclusions, References, and Acknowledgment.

The procedures for submission are as follows:

a. The student will submit the *Technical Paper* to the Supervisor/s and send a copy to the FYP Coordinator.
b. The FYP Coordinator will distribute one copy to the Examiners.
c. The Supervisor and Examiner will evaluate the *draft of the Technical Paper* and submit the marks to the FYP Coordinator.

5.10 Submission of Dissertation (Draft final report) (FYP 2)

Students should be able to explain in writing about the contents of the project and its significance, the problem statement, objectives, scope, literature review, methodology used, results, conclusions and recommendations in their dissertations.

The procedures for submission are as follows:

(a) Students are required to submit a *final report* to the Supervisor/s and send a copy to the FYP Coordinator.
(b) The FYP Coordinator will distribute one copy to the Examiners.
(c) The Supervisor and Examiners will evaluate the *draft final report* and submit the marks to the FYP Coordinator using **Form 14 as in Appendix 3-14** after the project presentation.

5.11 Project presentations (product demo, where applicable)

In this session, students should be able to:

(a) Verbally report the outcome of their final year project.
(b) Demonstrate how well they can explain and understand the project that they have been working on.
(c) Utilize their skills in oral presentation.

The procedures are as follows:

(a) Project presentation session(s) will be conducted face to face (online subject to department Chair's approval).
(b) The project presentation evaluation will be conducted (at a scheduled time) using **Form 13** in **Appendix 3-13**.

(c) The Supervisor/s and Examiners will give comments on the *final draft of the report* and the oral presentation of the project.

(d) Students must defend their findings in the reports and make necessary amendments as suggested by the Supervisors and Examiners before submitting the *final dissertation*.

(e) The presentation contents may focus on the following items:

- Problem Statement
- Objectives and Scope of Study
- Literature Review
- Procedure/Methodology
- Results and Findings
- Conclusion and Recommendation

(f) The Panel of Examiners comprise of the Supervisor, an external Examiner and an internal Examiner.

(g) The Supervisor and Examiners will evaluate the project presentation and submit the marks to the FYP Coordinator by using **Form 13 as in Appendix 3-13** after the project presentation sessions.

5.12 Submission of Hard-Bound Copy of Project Dissertation

(a) The student must submit two (2) hard copies—one for the Supervisor and one for personal keeping—and send one soft copy to the FYP Coordinator. The FYP Coordinator will upload the soft copy of the dissertation to UShare, in a folder designated for each department. The file must be submitted in a PDF format. Students who fail to submit the hard-bound dissertation will not receive their results transcript from the Examination Unit.

5.13 Grading of Project

(a) The FYP Coordinator will compile all the marks for the purpose of vetting at the department level.

(b) The FYP Coordinator will obtain endorsement of the FYP results from the FYP Chairperson after the results have been vetted by the FYP Committee at the department level.

(c) The FYP Coordinator will submit the endorsed result or grading to the Exam Unit as mentioned in **Section 8**.

6. WRITING FORMAT

Each interim report, technical paper, and dissertation must adhere to the specified formats. These types of academic writings comprise multiple sections that should be organized in a specific sequence. It is recommended that the contents of the student's scholarly work follow the order outlined below:

6.1 FYP 1 Interim Report

(a) Title Page

(b) Abstract

(c) Chapter 1: Introduction

- Background
- Problem Statement
- Objectives and Scope of Study

(d) Chapter 2: Literature Review and/or Theory

(e) Chapter 3: Methodology/Project Work

(f) Chapter 4: Conclusion and Future work

(g) References

(h) Appendices

6.2 Technical Paper

General guidelines for the technical paper preparation are as follows:

(a) Abstract: less than 300 words

(b) Keywords: 6 words

(c) Format:

- Times New Roman 11 / Arial 10
- 1 ½ spacing.

(d) Margin text

- Top: 1 in (2.5 cm)
- Bottom: 1 inch (2.5cm)
- Right: 1 inch (2.5cm)
- Top: 1 in (2.5 cm)
- Left: 1 inch (2.5 cm)

(e) One-column page writing

(f) Formula and Equation: MathType format

(g) Figures and tables: Labelled and embedded in the text.

(h) If authors are using databases, tables, diagrams, etc., from the internet and print materials, permission must be sought first from the rights holder. Please attach the permission(s) with the manuscript.

(i) Diagrams, tables, figures, and source file in jpeg or eps format must be attached separately.

(j) Reference style should be in IEEE Formatting Style.

(k) All references mentioned in the Reference List are cited in the text, and vice versa.

6.3 Dissertation

(a) Title Page

(b) Certification

(c) Abstract

(d) Acknowledgements

(e) Table of Contents

(f) List of Figures

(g) List of Tables

(h) Abbreviations and Nomenclatures

(i) Chapter 1: Introduction

- Background
- Problem Statement
- Objectives and Scope of Study

(j) Chapter 2: Literature Review and/or Theory

(k) Chapter 3: Methodology/Project Work

(l) Chapter 4: Results and Discussion

(m) Chapter 5: Conclusion and Recommendation

(n) References

(o) Appendices

6.4 General Writing Format

Students must follow specific guidelines for all academic writings as indicated in **Section 6**.

(a) Language

The dissertation must be written in acceptable and formal English. Use the passive voice.

(b) Font and Spacing

All text should be 1.5 spacing between lines and 3 spacing between paragraphs (Times New Roman regular font-style, size 12) typed on white A4 papers. The interim report should be in the form of double-sided printing. The hard-bound printed copy of the dissertation should be in the form of single sided printing.

The following however should be single spaced:

- Tables and figures
- Computer programs/source codes (must be reduced to font size 8)

(c) Length

The maximum length of the report, excluding appendices is as follows:

Interim Report	20 pages
Technical Paper	06 pages
Dissertation	50 pages

Students are encouraged to use brief and straightforward wordings, use passive voice and avoid using jargon as much as possible.

(d) Pagination

All pages must be numbered in proper sequence from introduction to the end of the report including pages on figures, tables, computer programs and appendices. All front materials are numbered in small Roman numerals (e.g. i, ii, iii). Page numbers appear by themselves and are not to be enclosed in parenthesis, hyphens or other decorative symbols. Page numbers must be positioned at the bottom and must be centred. Please refer to **Appendix 4-1**.

(e) Margin

The top, bottom and right margins are 25 mm except the left margin, which is 40 mm. Please refer to **Appendix 4-1**. All paragraphs should start from the left margin.

(f) Mathematical Equations

Mathematical equations must be spaced out; superscript and subscript must be clearly shown and numbered.

(g) Heading

The report should not have more than three levels of numbered headings as follows:

1. FIRST-LEVEL HEAD

1.1 Second-Level Head

1.1.1 Third-Level Head

All headings should be in Times New Roman and bold. Chapter and major headings should be in capitals and in 14 font size and 12 font size, respectively. Secondary and tertiary headings should be in title case and in 12 font size.

(h) Tables and Figures

Tables and figures are considered part of the report if it is within the main text. If it is of the size that is less than a page, it should be inserted into the text near the point of reference with a 3 spacing from the text. Tables should be on the same page. Margin limits of figures and tables should be the same as the full-page text. All tables and figures should be numbered consecutively. Table heading should be positioned at the top and centred. The numbers for figure should be positioned at the bottom and centred. Please refer to **Appendix 4.2.** Refer to each table or figure clearly in the text before placing it on the page. (For example, "Figure 1 shows ….)

(i) Documenting Sources

Students are required to cite the sources from which ideas were taken. Please refer to **Appendix 5-1**. **The documentation system to be used is the American Psychological Association (APA) or International Electrical Electronic Engineering (IEEE) format.**

(j) References

The method of writing references must follow the standard format. The sample reference format is in **Appendix 5-2**. This sample is using the APA and IEEE format.

(k) Title Page

The title page of the *interim report* and *dissertation* should be set out in accordance with the attached sample sheet in **Appendix 6-1** and should include the following:

- The title of the reports/ dissertation.
- The name of the candidate in FULL.
- The degree for which he/she is submitting the reports/ dissertation.

- The semester in which the reports/dissertation is submitted.
- The University name and address.

Students must submit **two (2) hard cover copies** of their dissertation to the FYP Coordinator. The hard cover colour for each programme is listed in **Table 1**. Please refer to **Appendix 6-2** for the writing format for the front hard cover.

Table 1: Hard cover colour for each Programme

Programme	**Colour**
Applied Chemistry	Purple
Applied Physics	Purple
Business Management	Navy Blue
Chemical Engineering	Dark Green
Civil Engineering	Black
Computer Engineering	Dark Brown
Electrical and Electronics Engineering	Dark Brown
Information Systems	Dark Blue
Information Technology	Dark Blue
Computer Science	Dark Blue
Materials Engineering	Maroon
Mechanical Engineering	Maroon
Petroleum Engineering	Dark Red
Petroleum Geoscience	Dark Green
Integrated Engineering	Royal Blue

6.5 General Content

This section will elaborate the general content needed in each part for each report format.

(a) Title Page

The title of the report should reflect the focus on core issues of the project work or issued related to it.

(b) Certification

This section is divided into two: certification of approval and certification of originality, as in **Appendix 7-1** and **7-2**. The certification of approval should be signed by the Supervisor after

he/she is satisfied with the corrections or amendments done by the student.

(c) Abstract

An abstract is a short version of a report. It covers the report's purpose, scope, methodology, results and conclusion. Abstracts should be no longer than one page as in **Appendix 8-1**.

(d) Acknowledgements

Acknowledgements should include the names of the contributors to the project work, including the Supervisors and the members of the group, preferably not more than one page.

(e) Table of Contents

Table of Contents lists all headings and sub-headings, tables, figures, appendices and, bibliography with page numbers. It also includes the certification, abstract and acknowledgement (if applicable). Please refer to the sample in **Appendix 9-1**.

(f) Introduction

The Introduction must include the background of the project, the problem statement, the objective(s) and scope of the study. The Problem statement needs to focus on the situation of the problem and research questions which lead to the objective(s) of the study. Students are required to clarify the boundary of the project work to ensure its feasibility within the given time frame.

(g) Literature Review and/or Theory

The Literature Review is the analytical, critical and objective review of written materials on the chosen topic and area. It provides the background information on the research question and identifies what others have said and/or discovered about the question. It contains all relevant theories, hypotheses, facts and data which are relevant to the objective and findings of the project.

(h) Methodology/Project Work

The Methodology refers to methods/procedures used by the student to achieve the objective(s) of the project. The methods/procedures must be relevant and acceptable.

(i) Results and Discussion

This section presents the findings or outcomes of the project work. All gathered data from the project work must be presented in the

form of tables and figures such as graphs, diagrams or appropriate formats. The data needs to be analysed, and the results need to be discussed.

(j) Conclusion and Recommendation

The Conclusion highlights the most significant findings in relation to the objective(s) of the project. This section should also include recommendations for future project work.

(k) References

This section is the list of references used in the project. The method of writing references must follow the standard format. Please refer to the sample reference format in **Appendix 5-2**.

(l) Appendices

Lengthy calculations, figures, raw data, computer programs/source codes, outputs, etc. are to be enclosed as appendices. They should be titled and numbered in chronological order and capital letters. The appendices and their titles need to be listed in the Table of Contents. Provide title for each appendix, for example "Appendix 1. Questionnaire Sample".

7. Evaluation

Students are evaluated based on their capability in undertaking the project, producing the written report and presenting the results. Overall commitment, as well as personal conduct, is also to be observed at all times. The main components of evaluation and the grading structure are given in **Table 2.1 and Table 2.2**.

Table 2.1: Grading Structure for FYP 1

Assessment (FoE & FSIT)	**Assessment Contribution (%)**	
	Supervisor	**Panel of Examiner(s)**
Progress assessment 1	10	-
Proposal defence	10	20
Progress assessment 2	10	-
FYP 1 Interim Report	20	30
Total	**50**	**50**

Table 2.2: Grading Structure for FYP 2

Assessment (FoE & FSIT)	Assessment Contribution (%)		
	Supervisor	Internal Examiner(s)	External Examiner(s)
Progress Assessment 1 (Professional & Ethics)	5	-	-
Progress Assessment 2 (Project Mngt)	5	-	-
Technical Paper	10	10	-
Project Presentation	5	10	15
Dissertation	15	15	10
Total	40	35	25
Overall Total	**40**	**60**	

8. GRADING SCHEME

Students will be graded according to the UTP grading scheme as in **Table 3**.

Table 3: UTP Grading Scheme

Score	Grade	Point
85 – 100	A	4.0
80 – 84.9	A-	3.75
75 – 79.9	B+	3.5
65 -74.9	B	3.0
55 – 64.9	C+	2.5
50 – 54.9	C	2.0
45 – 49.9	D+	1.5
40 – 44.9	D	1.0
0 - 39.9	F	0.0

9. CLAIMS

Students are entitled to claim the total of RM 500 for the whole Final Year Project. The claimed amount must had been spent on any of the following items.

(a) Consumables and equipment

(b) Testing/technical support

(c) Consultation or other support

(d) Travel/visit for data requisition (with the most economical mode of travel)

(e) Procurement of data

* Students are not entitled to claim for the cost of printing hard-bound thesis

All claims shall be submitted using **Form 03** (with the Official receipts/Invoice) to the Supervisor, requires support from the FYP Coordinator and must be endorsed by the FYP Chairperson.

10. PLAGIARISM

Plagiarism is the act of obtaining or attempting to obtain credit for academic work by representing the work of another as one's own without the necessary and appropriate acknowledgment. More specifically, plagiarism is:

(a) The act of incorporating the ideas, words of sentences, paragraphs, or parts thereof without appropriate acknowledgment and representing the product as one's own work; and

(b) The act of representing another's intellectual work such as musical composition, computer program, photographs, painting, drawing, sculpture, or research or the likes as one's own.

(Source : www.sonoma.edu/uaffairs/policies/cheatingpolicy.htm)

If a student is in doubt of the nature of plagiarism, he/she should discuss the matter with the Supervisor. If a student is caught committing plagiarism, stern action will be taken against the student. This includes the student being given zero marks for the assessment in FYPI. However, for FYPII, the student will be given a grade of F.

11. COPYRIGHT

The university shall be the owner for all findings, designs, patents, and other intellectual property rights.

MAPPING OF SUBJECT – PROGRAM OUTCOME, COURSE LEARNING OUTCOME AND ASSESSMENT TOOLS FOR FYP1 AND FYP2

Faculty of Engineering

PO	PO1 (C)	PO2 (C)	PO3 (C)	PO4 (P)	PO5 (C)	PO6 (A)	PO7 (A)	PO8 (A)	PO9 (A)	PO10 (A)	PO11 (A)	PO12 (A)
FYP 1		X						X		X		X
FYP 2				X	X			X		X	X	

FYP1		PO2 (C)	PO8 (A)	PO10 (A)	PO12 (A)	Assessment Tools
CLO1	Formulate research problem through critical literature review				X	Progress assessment 1
CLO2	Propose appropriate methodology to solve the research problem	X				Interim Report
CLO3	Communicate the proposed research project through effective presentation and report			X		Proposal Defense, Interim Report
CLO4	Adapt to professional and ethical responsibilities in research work undertaking		X			Progress assessment 2

FYP2		PO4 (P)	PO5 (C)	PO8 (A)	PO10 (A)	PO11 (A)	Assessment Tools
CLO1	Analyse and interpret data using appropriate techniques / tools to solve complex research problems with understanding of its limitation	X	X				Dissertation (40%)
CLO2	Demonstrate professional and ethical responsibilities in project undertakings			X			Progress assessment 1 (5%)
CLO3	Communicate the deliverables of the research project effectively, in verbal and written form				X		Project presentation (30%), Technical paper (20%)
CLO4	Demonstrate research project management effectively					X	Progress assessment 2 (5%)

MAPPING OF SUBJECT – PROGRAM OUTCOME, COURSE LEARNING OUTCOME AND ASSESSMENT TOOLS FOR FYP1 AND FYP2

Faculty of Science and Information Technology

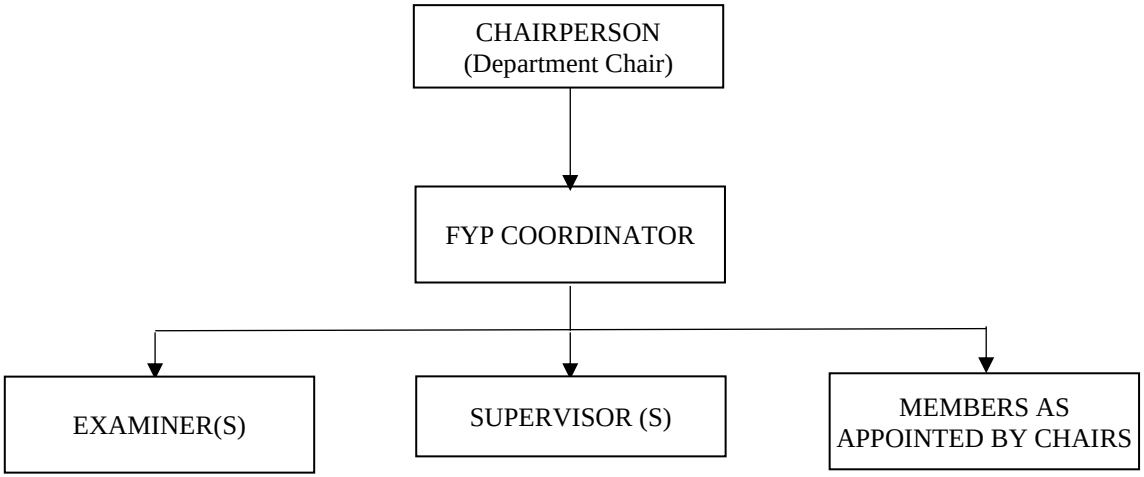
PO (Cluster)	1	2	3a	3d	3f	5	3b	3c
FYP 1		X				X		X
FYP 2			X		X	X	X	X

FYP1		PO (Cluster)				Assessment Tools
		2 (C)	3b+c (A)	4 (A)	5 (A)	
CLO1	Formulate research / problem through critical literature review			X		Progress assessment 1
CLO2	Propose appropriate methodology / conceptual framework for the research / project	X				Interim Report
CLO3	Communicate the proposed research / project through effective presentation and report		X			Proposal Defense, Interim Report
CLO4	Adapt to professional work ethic and practices in research / project work undertaking				X	Progress assessment 2

FYP2		PO (Cluster)				Assessment Tools
		3a	3b+c	3f	5	
		PLO3	PLO4	PLO6	PLO8	
CLO1	Analyse and interpret data using appropriate techniques / tools to solve the identified research / project problem with understanding of its limitation	X				Dissertation (32.5%) Technical Paper (10%)
CLO2	Adapt to professional and ethical responsibilities in research / project work undertaking		X			Progress assessment 1 (5%)
CLO3	Communicate effectively in verbal and written form in reporting the research / project outcome			X		Project Presentations (30%), Dissertation (7.5%), Technical paper (10%)
CLO4	Undertake research / project management effectively				X	Progress assessment 2 (5%)

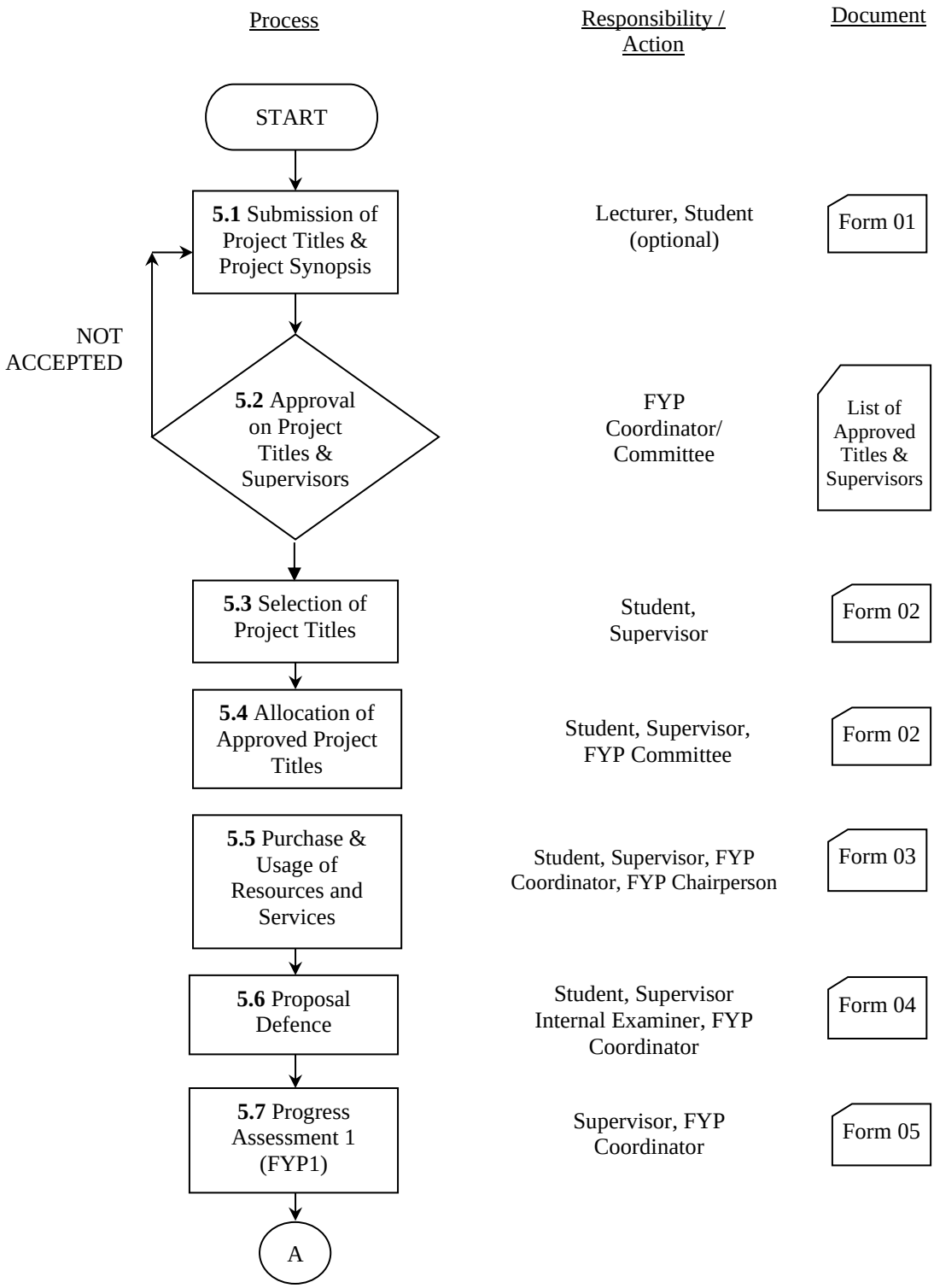
APPENDIX 1-2

Organisation Chart for FYP Committee



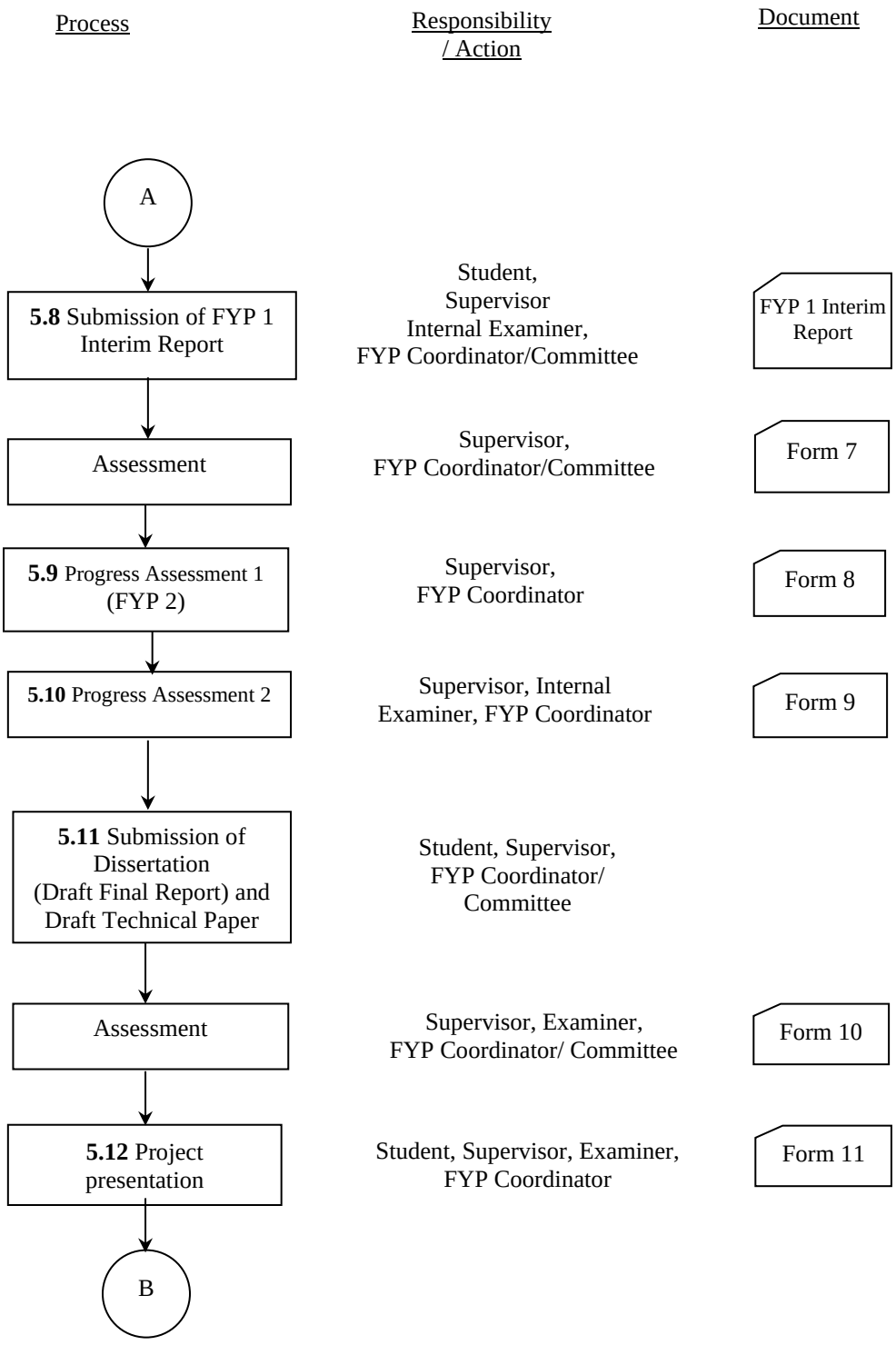
APPENDIX 1-3

FINAL YEAR PROJECT FLOW PROCESS



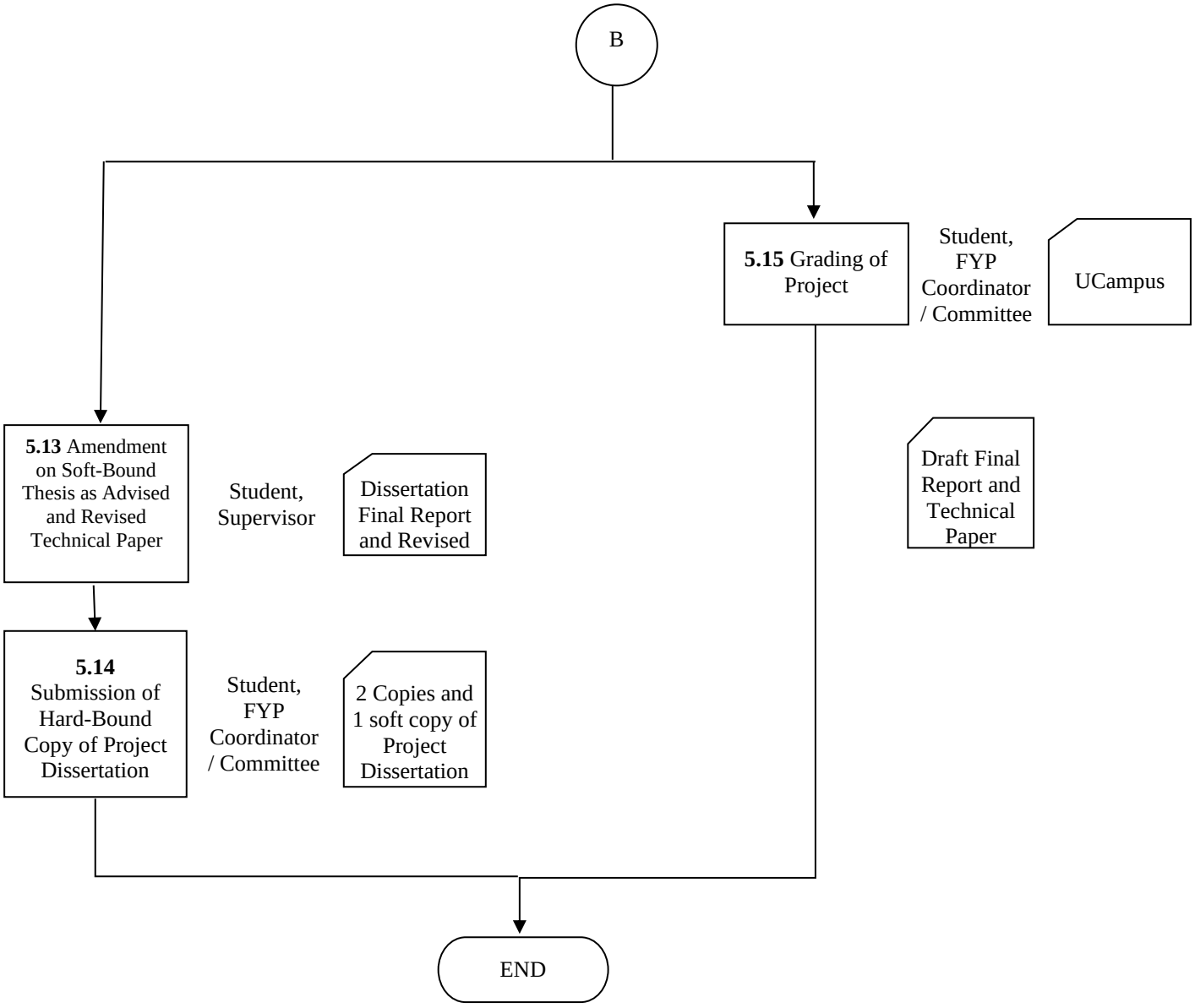
APPENDIX 1-3

FINAL YEAR PROJECT FLOW PROCESS



APPENDIX 1-3

FINAL YEAR PROJECT FLOW PROCESS



Timelines for FYP 1

No.	Detail/ Week	-1	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	Selection of Project Topic[a]	■	■													
2	Preliminary Research Work		■	■	■	■	■	■	■	■	■	■	■	■		
3	Submission of Progress Assessment 1[b]							●								
4	Proposal Defence								■	■						
5	Submission of Interim Draft Report												●			
6	Submission of Progress Assessment 2[b]													●		
7	Submission of Interim Report													●		

● Suggested milestone

■ Process

Note:

a. The process of topic selection should start prior to semester commencement. Topics should be finalised and assigned latest by Week 1 of the semester.

b. Item 3 and Item 6 is a continuous assessment by SV. Submission of Progress Assessment 1 & 2 is an action item for SV.

Timelines for FYP 2

No.	Detail/ Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	Project Work Continues	■	■	■	■	■	■	■	■	■	■	■	■			
2	Submission of Progress Assessment 1[a]						●									
3	Submission of Draft Dissertation and Draft Technical Paper									●						
4	Submission of Dissertation (soft bound) and Revised Technical Paper										●					
5	Project Presentation												●			
6	Submission of Progress Assessment 2[a]												●			
7	Submission of Project Dissertation (Hard Bound)														●	

● Suggested milestone

■ Process

Note:

a. Item 2 and Item 6 is a continuous assessment by SV. Submission of Progress Assessment 1 & 2 is an action item for SV.
b. Draft thesis as formative assessment, a pre-requisite to submit final thesis. Failure to submit draft means students will be bar from submitting final thesis.

Doc. Ref. No.	Issue Version	Date
UTP-ACA-PROG-FYP-01.01a	7.0	Jan 2025

FORM 01A

UNIVERSITI TEKNOLOGI PETRONAS

FINAL YEAR PROJECT TITLE PROPOSAL

Project Title:

Proposer's Name :_____Student's ID:_____

Proposer's e-mail address: _____H/P No: _____

CGPA: _____Total Credit Hours This Semester: _____

Area / specialization: _____

Suggested Supervisor (if any) : _____

Collaborator(s) (if any) : _____

Problem Statement:

_____

_____

_____

Objectives:

_____

_____

Pre-requisite (if any):

_____

_____

Short summary of the research project:

Tools/equipment required:

APPENDIX 3-1b

Doc. Ref. No.	Issue Version	Date
UTP-ACA-PROG-FYP-01.01b	7.0	Jan 2025

UNIVERSITI TEKNOLOGI PETRONAS

FORM 01B

FINAL YEAR PROJECT TITLE PROPOSAL

Project Title:

Supervisor's Name: _____

Area / specialization: _____

Collaborator(s) (if any): _____

Co-Supervisor (s) (if any): _____

Project Background:

_____

_____

Problem Statement:

_____

_____

_____

Objectives:

_____

_____

Tools/equipment required:

_____

_____

_____

Project Deliverables:

_____

_____

_____

UNIVERSITI TEKNOLOGI PETRONAS

Doc. Ref. No.	Issue Version	Date
UTP-ACA-PROG-FYP-01.02	7.0	Jan 2025

FORM 02

TOPIC SELECTION

(To be completed by student in 2 copies.)

Semester : _____Year: _____

Name : _____

Student's ID : _____ Programme: _____

List of topic chosen (ranking in order of preference):

	Proj. No.	**Proj. Title**	**Supervisor**
1			
2			
3			

Justification for the chosen topics:

_____

_____

_____

_____

Students should return this form at least one week before the start of the semester to the respective department Final Year Project Coordinator

COMMITTEE DECISION

Topic approved: _____

Approved by FYP Coordinator *(only for Supervisor-allocated titles)*	
Signature:	
Official stamp:	
Date:	

UNIVERSITI TEKNOLOGI PETRONAS

Doc. Ref. No.	Issue Version	Date
UTP-ACA-PROG-FYP-01.03	7.0	Jan 2025

FORM 03

FINAL YEAR PROJECT REQUISITION FORM

(Note: Supervisor can use this form to request (a) purchasing consumables and equipment, (b) testing / technical support, (c) consultations fees and other related resources to the project, (d) visit for data collection.)

REQUESTED BY	
Supervisor's Name:	
Project Title:	
Supervisor's Signature:	
Student's Name:	
Date:	

No.	Description	Purpose	Quantity	Estimated Cost

ENDORSEMENT BY CO-ORDINATOR/ LABORATORY EXECUTIVE		**APPROVAL BY COMMITTEE CHAIRPERSON**	
Name		Name	
Programme		Programme	
Signature		Signature	
Date		Date	
Comment:		Approval ☐ Approve ☐ Not Approve Comment:	

** Please attach this form with original receipt or any proof of purchase.*

** Students are not entitled to claim for the cost of printing hard-bound thesis*

UNIVERSITI TEKNOLOGI PETRONAS

FYP I – PROGRESS ASSESSMENT 1

(To be completed by Supervisor)

Student's Name : _____ Student's ID: _____ Programme : _____

Project Title : _____

Evaluation criteria	Very Weak (1)	Weak (2)	Fair (3)	Good (4)	Very Good (5)	Multiplier	Marks
Interest	Lack of interest in exploring issues for a given task	Demonstrate limited interest in exploring issues for a given task	Demonstrate moderate interest in exploring issues for a given task	Demonstrate good interest in exploring issues for a given task	Demonstrate excellent interest in exploring issues for a given task	**x 1**	
Initiative	Lack of initiative to complete a task	Demonstrate limited initiative to complete a task	Demonstrate moderate initiative to complete a task	Demonstrate good initiative to complete a task	Demonstrate excellent initiative to complete a task	**x 1**	
Effort	Lack of effort to complete task	Minimal effort to complete task	Moderate effort to complete task	Good effort to complete task	Excellent effort to complete task	**x 1**	
Self-Learning	• Not able to self-learn • Did not understand project and lack of self-initiative in handling and planning the task for the project	• Limited ability to self-learn • Have low understanding on the project and show minimal self-initiative in handling and planning of the tasks for the project	• Moderate ability to self-learn • Understood some part of the project and show moderate self-initiative in in handling and planning of the tasks for the project	• Good ability to self-learn • Understood most part of the project and show good self-initiative in handling and planning of the tasks for the project	• Excellent ability to self-learn • Understood the project well and show excellent self-initiative in in handling and planning of the tasks for the project	**x 1**	
						Total Score	**/ 20**

Comments:

_____

_____

……..…

Supervisor's Signature

Name:

Date:

UNIVERSITI TEKNOLOGI PETRONAS

FYP I – PROPOSAL DEFENCE

(To be completed by Supervisor/ Examiner)

Student's Name : _____ Student's ID: _____ Programme : _____

Project Title : _____

Evaluation criteria	Very Weak (1)	Weak (2)	Fair (3)	Good (4)	Very Good (5)	Multiplier	Marks
Subject knowledge and content	Not able to present research ideas* and its implementation clearly and require major improvements	Able to present research ideas* and its implementation with limited clarity and require further improvements	Able to present research ideas* and implementation fairly clearly and require minor improvements	Able to present the research ideas* professionally and clearly, with some concern on its implementation	Able to present the research ideas* and its implementation professionally and clearly, showing mastery of the materials.	**x 5**	
Confident delivery of research ideas	Not able to deliver research ideas confidently, with poor language choice and frequent use of filler words	Able to deliver research ideas with limited confidence and occasional use of filler words, thus requiring major improvements	Able to deliver research ideas with fair confidence, with little use of filler words, and require minor improvements	Able to deliver research ideas confidently, with good language choice and tone	Able to deliver research ideas with excellent confidence	**x 1**	
Effective and articulate delivery of research ideas	Not able to deliver research ideas effectively	Able to deliver research ideas with limited effect and require major improvements	Able to deliver research ideas fairly effectively and require minor improvements	Able to deliver research ideas effectively and articulately	Able to deliver research ideas with great effect and articulate	**x 1**	
Use of visual aids and supportive materials	Not able to use visual aids / supporting materials effectively (disorganised slides, unreadable materials)	Able to use visual aids / supporting materials with minimum impact, requiring major improvement	Able to use visual aids / supporting materials that shows general understanding of the research ideas	Able to use visual aids / supporting materials that shows enhanced understanding of the research ideas.	Able to use visual aids / supporting materials that significantly support the research ideas and shows excellent understanding on the research ideas	**x 1**	
Response to questions	Not able to understand and respond to a question	Able to understand and answer questions but not able to accurately answer the question	Able to understand and answer questions satisfactorily	Able to understand and answer questions well	Able to fully understand and respond to questions extremely well	**x 1**	
						Total Score	**/ 45**

UNIVERSITI TEKNOLOGI PETRONAS

Doc. Ref. No.	Issue Version	Date
UTP-ACA-PROG-FYP-01.05b	7.0	Jan 2025

APPENDIX 3-5

FORM 05

FYP I – PROPOSAL DEFENCE

Comments:

……..…

Supervisor's / Examiner's Signature

Name:

Date:

UNIVERSITI TEKNOLOGI PETRONAS

FYP I – PROGRESS ASSESSMENT 2

(To be completed by Supervisor)

Student's Name : _____ Student's ID: _____ Programme : _____

Project Title : _____

Evaluation criteria	Very Weak (1)	Weak (2)	Fair (3)	Good (4)	Very Good (5)	Multiplier	Marks
Work Ethics	Practice inappropriate working culture such as bad behavior, no punctuality as well as not being efficient, productive and ethical at work in all situations	Practice less appropriate working culture such as inconsistent behavior, less punctuality as well as being less efficient, productive and ethical at work in many situations	Practice good working culture such as good moral, timeliness as well as being efficient, productive and ethical at work in general	Practice good working culture such as good morality, timeliness as well as being efficient, productive and ethical at work in most situations	Always practice excellent working culture such as good morality, timeliness as well as being efficient, productive and ethical at work in all situations	**x 1**	
Integrity	Perform a task with lack of trust, honesty, sincerity and transparency	Perform a task with limited trust, honesty, sincerity and transparency	Perform a task with acceptable trust, honesty, sincerity and transparency	Perform a task with trust, honesty, sincerity and transparency in most situations	Always perform a task with trust, honesty, sincerity and transparency in any situation	**x 1**	
Work Responsibility	Does not perform assigned tasks within the scope of work even with close supervision	Perform assigned tasks within the scope of work with close supervision	Perform assigned tasks within the scope of work and meet expectation	Perform assigned tasks within the scope of work and exceed expectation	Perform assigned tasks beyond the scope of work and exceed expectation	**x 1**	
						Total Score	**/ 15**

Comments:

_____

_____

……..…

Supervisor's Signature

Name:

Date:

UNIVERSITI TEKNOLOGI PETRONAS

FYP I – PROGRESS MONITORING (To be used together with Form 04 and Form 06)

(To be completed by Supervisor)

PROGRESS MONITORING REPORT			
Project title			
Student's Name		Student's ID	
Supervisor's Name		Programme	

A. PROJECT STATUS

No.	**Task / Milestone**	**Progress status**					
		Progress Meeting 1	Progress Meeting 2	Progress Meeting 3	Progress Meeting 4	Progress Meeting 5	Progress Meeting 6

UNIVERSITI TEKNOLOGI PETRONAS

FYP I – PROGRESS MONITORING (To be used together with Form 04 and Form 06)

B. RISK MANAGEMENT – Please list the identified risk that will delay the project progress, and its mitigation plan.
Student's Signature: Date:

Comments:

_____

_____

_____

……..…
Supervisor's Signature
Name:
Date:

UNIVERSITI TEKNOLOGI PETRONAS

FYP I – INTERIM REPORT

(To be completed by Supervisor/ Examiner)

Student's Name : _____ Student's ID: _____ Programme: _____

Project Title : _____

PART A

Evaluation criteria		Very Weak (1)	Weak (2)	Fair (3)	Good (4)	Very Good (5)	Multiplier	Marks
ABSTRACT (5)	• Problem statement • Objective • Methodology • Future work	Abstract is irrelevant and does not provide the summary of the whole project	Abstract is relevant but did not include some of the elements stated	Abstract is relevant, and include all elements, however, is vague and fail to highlight appropriate details about the research	Abstract is relevant, include all elements, and highlight appropriate details about the research.	Abstract is relevant, clear and concise description of all elements, highlighting the importance of the research.	**x 1**	
INTRODUCTION (25)	Background of Study and problem statement	Irrelevant background of study and problem statement	Background of study is partially relevant, and unclear problem statement	Background of study is sufficient to highlight the research work, however, problem statement is generic and unclear	Good background of study that shows connection to the problem statement. Problem statement is clearly presented.	Well written, articulate and concise background of study that justifies the problem statement. Problem statement is clearly presented.	**x 2**	
	Objective	Research objectives are irrelevant to the problem statement. Unachievable within the time frame	Research objectives are unclear and partially relevant to the problem statement. Unachievable within the time frame	Research objectives are clear, however, partially relevant to problem statements. Achievable within the time frame	Research objectives are clear, relevant to problem statement and achievable within the time frame	Research objectives are clearly stated and justified towards answering the problem statement, and achievable within the time frame	**x 2**	
	Scope of Study, including relevancy and feasibility	Scope of Study are irrelevant to research objectives	Scope of Study are partially relevant to research objectives	Scope of Study are relevant to research objectives but poorly described	Scope of Study are relevant to research objectives and fairly described	Scope of Study are relevant to research objective and clearly stated, highlighting the feasibility of the work within the time frame	**x 1**	
Evaluation criteria		**Very Weak (1)**	**Weak (2)**	**Fair (3)**	**Good (4)**	**Very Good (5)**	**Multiplier**	**Marks**

FYP I – INTERIM REPORT

LITERATURE REVIEW (30)	**Critical analysis, relevancy and recentness of literature**	Irrelevant literature review	Literature review explains previous study, without connecting it to the current study. Lack in critical analysis.	Literature review explains previous study relevant to the current study, however, lack in critical analysis.	Literature review provides good discussion on previous study relevant to the current study, and sufficiently highlights the problem statement.	Literature review is comprehensive, critically analyse previous work relevant to current studies, with proper cross referencing. Clearly established the importance of the current work.	**x 4**	
	Quality of references (journals/conferences/etc.)	Insufficient references, none from published journal / conferences	Insufficient references, minimal number of references from published journal / conferences	Adequate references, some are relevant, and from published journal / conferences.	Sufficient references, almost all are relevant, and almost all are from published journal / references	Sufficient references all are relevant, and almost all from published journal / references	**x 2**	
METHODOLOGY (15)	Research methodology and Project Activities	Insufficient and poorly described methodology and project activities	Methodology and project activities are unsuitable to achieve the objectives	Methodology and project activities are partially relevant to achieving the objectives	Methodology and project activities are relevant to achieving the objectives and fairly described and achievable within the time frame	Methodology and project activities are relevant to achieving the objectives and comprehensively described, and highly achievable within the time frame	**x2**	
	Project milestone and timeline (Gantt-Chart) (for both semesters)	No Project timeline (Gantt-Chart)	Unclear, unfeasible and unstructured study plans	Moderately clear, feasible but unstructured study plans	Clear, feasible and structured study plans	Exceptionally clear, feasible and very well-structured study plans	**x1**	
CONCLUSION (10)	**Summary of project progress and future work**	Insufficient conclusion and irrelevant with objective. No future work recommended	Insufficient conclusion and irrelevant with objective. Insufficient future work recommended	Sufficient conclusion and sufficiently relevant with objective. Sufficient future work recommended	Provide good conclusion and recommendation for future work, relevant to the objective.	Provide excellent conclusion that is comprehensive, and recommended future work is relevant to objective.	**x 2**	
							Total Score	**/ 85**

FYP I – INTERIM REPORT

PART B

Evaluation criteria		Very Weak (1)	Weak (2)	Fair (3)	Good (4)	Very Good (5)	Multiplier	Marks
ORGANISATION AND CLARITY (15)	Standard Guidelines and formatting	Report does not follow the standard guidelines and formatting	Report that has frequent major mistakes with the standard guidelines and format	Report that has frequent minor mistakes with the standard guidelines and format	Report that has some mistakes with the standard guidelines and format	Report that complies with the standard guidelines and format	**x 1**	
	English usage and writing skills (writing skills refers to clear, coherent and systematic writing of ideas)	Very poor English and writing skills.	Poor English and writing skills with many spellings and grammar mistake	Fair English and writing skills with noticeable spelling and grammar mistake	Good English and writing skills with minimal spelling or grammar mistake	Excellent English and writing skills with no spelling or grammar mistakes	**x 1**	
	Referencing and citations	Most reference are missing from reference list and did not follow style guide	Many references are missing from reference list and did not follow style guide	Some references are missing from reference list and / or most did not follow style guide	Most references are cited in the reference list, which follow the style guide	All references are cited in the reference list, which follow the style guide	**x 1**	
							Total Score	**/ 15**

Comments:

_____

_____

……..…

Supervisor's / Examiner's Signature

Name:

Date:

UNIVERSITI TEKNOLOGI PETRONAS

FYP II – PROGRESS ASSESSMENT 1

(To be completed by Supervisor)

Student's Name : _____ Student's ID: _____ Programme: _____

Project Title : _____

Evaluation criteria	Very Weak (1)	Weak (2)	Fair (3)	Good (4)	Very Good (5)	Multiplier	Marks
Work Ethics	Practice inappropriate working culture such as bad behavior, no punctuality as well as not being efficient, productive and ethical at work in all situations	Practice less appropriate working culture such as inconsistent behavior, less punctuality as well as being less efficient, productive and ethical at work in many situations	Practice good working culture such as good moral, timeliness as well as being efficient, productive and ethical at work in general	Practice good working culture such as good morality, timeliness as well as being efficient, productive and ethical at work in most situations	Always practice excellent working culture such as good morality, timeliness as well as being efficient, productive and ethical at work in all situations	**x 1**	
Integrity	Perform a task with lack of trust, honesty, sincerity and transparency	Perform a task with limited trust, honesty, sincerity and transparency	Perform a task with acceptable trust, honesty, sincerity and transparency	Perform a task with trust, honesty, sincerity and transparency in most situations	Always perform a task with trust, honesty, sincerity and transparency in any situation	**x 1**	
Work Responsibility	Does not perform assigned tasks within the scope of work even with close supervision	Perform assigned tasks within the scope of work with close supervision	Perform assigned tasks within the scope of work and meet expectation	Perform assigned tasks within the scope of work and exceed expectation	Perform assigned tasks beyond the scope of work and exceed expectation	**x 1**	
						Total Score	**/ 15**

Comments:

_____

_____

……..…

Supervisor's Signature

Name:

Date:

UNIVERSITI TEKNOLOGI PETRONAS

FYP II – PROGRESS ASSESSMENT 2

(To be completed by Supervisor)

Student's Name : _____ Student's ID: _____ Programme: _____

Project Title : _____

Evaluation criteria		Very Weak (1)	Weak (2)	Fair (3)	Good (4)	Very Good (5)	Multiplier	Marks
PROJECT MANAGEMENT	Progress discussion and time management	• Meet Supervisor less than 4 times during the semester • Never show any improvement on research / project progress during discussion with Supervisor • Always late for meeting	• Meet Supervisor less than 4 times during the semester • Shows very little improvement on research / project progress during discussion with Supervisor • Almost always late for meeting	• Meet Supervisor at least 5 times during the semester • Shows moderate improvement on research / project progress during discussion with Supervisor • Sometimes late for meeting	• Student meets Supervisor at least 7 times during the semester • Student shows improvement as per project timeline on research / project progress during discussion with Supervisor • Punctual most of the time	• Student meets Supervisor more than 7 times during the semester • Student shows improvement as per project timeline as well as able to constructively discuss the research / project progress during discussion with Supervisor • Always punctual	**x 2**	

UNIVERSITI TEKNOLOGI PETRONAS

FYP II – PROGRESS ASSESSMENT 2

Evaluation criteria		Very Weak (1)	Weak (2)	Fair (3)	Good (4)	Very Good (5)	Multiplier	Marks
	Project implementation	• Research / project work does not follow the proposed methodology • Less than 20% key milestone achieved within timeline	• Research / project work follows the proposed methodology with major modification • 40% key milestone achieved within timeline	• Research / project work follows the proposed methodology with minor modification • 60% key milestone achieved within timeline	• Research / project work follows the proposed methodology • 80% of key milestone achieved within timeline	• Research / project work follows the proposed methodology • All key milestone achieved within timeline	**x 1**	
	Risk Management	Unable to identify any risk that may delay the project completion	Risk identified, but no action taken to mitigate risk	Risk identified, and 50% of risk are mitigated	Risk identified, and 80% of risk are mitigated	Risk identified, and all risk are mitigated	**x 1**	
							Total Score	**/ 20**

Comments:

_____

_____

……..…

Supervisor's Signature

Name:

Date:

UNIVERSITI TEKNOLOGI PETRONAS

FYP II – PROGRESS MONITORING (To be used together with Form 09 & Form 10)

(To be completed by Supervisor)

PROGRESS MONITORING REPORT			
Project title			
Student's Name		Student's ID	
Supervisor's Name		Programme	

A. PROJECT STATUS

No.	**Task / Milestone**	**Progress status**					
		Progress Meeting 1	Progress Meeting 2	Progress Meeting 3	Progress Meeting 4	Progress Meeting 5	Progress Meeting 6

UNIVERSITI TEKNOLOGI PETRONAS

FYP II – PROGRESS MONITORING (To be used together with Form 09 & Form 10)

B. RISK MANAGEMENT – Please list the identified risk that will delay the project progress, and its mitigation plan.
Student's Signature: Date:

Comments:

_____

_____

……..…

Supervisor's Signature

Name:

Date:

UNIVERSITI TEKNOLOGI PETRONAS

FYP II – PRACTICAL SKILLS (FOR FACULTY OF SCIENCE & INFORMATION TECHNOLOGY ONLY)

(To be completed by Supervisor)

Student's Name : _____ Student's ID: _____ Programme : _____

Project Title : _____

Evaluation criteria		Very Weak (1)	Weak (2)	Fair (3)	Good (4)	Very Good (5)	Multiplier	Marks
PRACTICAL SKILLS	**Originality**	Lack of original ideas to solve the research problem	Weak idea to solve the research problem	Moderate idea to solve the research problem	Good original idea to solve the research problem	Excellent new and original idea to solve the research problem	**x 1**	
	Critical observation	Lack of attention to research activities	Limited attention to research activities	Moderate attention to research activities	Good attention to research activities	Mindful of research activities	**x 1**	
	Critical thinking	Lack of analysis of results to form conclusions	Limited analysis of results to form conclusions	Moderate analysis of results to form conclusions	Acceptable analysis of results to form conclusions	Clear and extensive analysis of results to form conclusions	**x 1**	
	Curiosity	Lack of curiosity/motivation to complete the project	Demonstrate limited curiosity/motivation to complete the project	Moderate curiosity/motivation to complete the project	Good curiosity/motivation in completing the project	Excellent curiosity/motivation in completing the project	**x 1**	
							Total Score	**/ 20**

Comments:

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……..…

Supervisor's Signature

Name:

Date:

UNIVERSITI TEKNOLOGI PETRONAS

FYP II – PROJECT PRESENTATION

(To be completed by Supervisor/ Examiner)

Student's Name : _____ Student's ID: _____ Programme: _____

Project Title : _____

Evaluation criteria	Very Weak (1)	Weak (2)	Fair (3)	Good (4)	Very Good (5)	Multiplier	Mark
Subject knowledge and content	• Not able to present research outcome* clearly • Presentation contains multiple fact errors; some are major in nature	• Able to present research outcome* with limited clarity • The presentation contains multiple fact errors	• Able to present research outcome* clearly • Presentation contains minimal fact errors	• Able to present the research outcome* professionally and clearly • Major facts are accurate	• Able to present the research outcome* professionally and clearly, showing mastery of the materials. • Facts presented are accurate	**x 5**	
Confident delivery of research ideas	Not able to deliver research outcome* confidently, with poor language choice and frequent use of filler words	Able to deliver research outcome* with limited confidence and occasional use of filler words, thus requiring major improvements	Able to deliver research outcome* with moderate confidence, with little use of filler word, thus require minor improvements	Able to deliver research outcome* confidently, with good language choice and tone	Able to deliver research outcome* with excellent confidence	**x 1**	
Effective and articulate delivery of research ideas	Not able to deliver research outcome* effectively	Able to deliver research outcome* with limited effect and require major improvements	Able to deliver research outcome* with moderate effect and require minor improvements	Able to deliver research outcome* effectively and articulately	Able to deliver research outcome* with great effect and articulate	**x 1**	

UNIVERSITI TEKNOLOGI PETRONAS

FYP II – PROJECT PRESENTATION

Evaluation criteria	Very Weak (1)	Weak (2)	Fair (3)	Good (4)	Very Good (5)	Multiplier	Mark
Use of visual aids and supportive materials (including product demo, if applicable)	Not able to use visual aids / supporting materials effectively (disorganised slides, unreadable materials)	Able to use visual aids / supporting materials with minimum impact, requiring major improvement	Able to use visual aids / supporting materials that shows general understanding of the research outcome*	Able to use visual aids / supporting materials that shows good understanding of the research outcome*.	Able to use visual aids / supporting materials and shows excellent understanding on the research outcome*	**x 1**	
Response to questions	Not able to understand and respond to a question	Able to understand and answer questions but not able to accurately answer the question	Able to understand and answer questions satisfactorily	Able to understand and answer questions well	Able to fully understand and respond to questions extremely well	**x 1**	
						Total Score	**/ 45**

Comments:

_____

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……..…

Supervisor's / Examiner's Signature

Name:

Date:

UNIVERSITI TEKNOLOGI PETRONAS

FYP II – DISSERTATION

(To be completed by Supervisor/ Examiner)

Student's Name : _____ Student's ID: _____ Programme: _____

Project Title : _____

PART A

Evaluation criteria		Very Weak (1)	Weak (2)	Fair (3)	Good (4)	Very Good (5)	Multiplier	Marks
Abstract (10)	• Problem statement • Objective • Methodology • Future work	Abstract is irrelevant and does not provide the summary of the whole project	Abstract is relevant but did not include some of the elements stated	Abstract is relevant, and include all elements, however, is vague and fail to highlight appropriate details about the research	Abstract is relevant, include all elements, and highlight appropriate details about the research.	Abstract is relevant, clear and concise description of all elements, highlighting the importance of the research.	**x 2**	
Introduction (10)	• Background of study and problem statement	Irrelevant background of study and problem statement	Background of study is partially relevant, and unclear problem statement	Background of study is sufficient to highlight the research work, however, problem statement is generic and unclear	Good background of study that shows connection to the problem statement. Problem statement is clearly presented.	Well written, articulate and concise background of study that justifies the problem statement. Problem statement is clearly presented.	**x 1**	
	• Objective and scope of study	• Research objectives are irrelevant to the problem statement. • Scope of Study are irrelevant to research objectives	• Research objectives are unclear and partially relevant to the problem statement. • Scope of Study are partially relevant to research objectives and poorly described	• Research objectives are clear, however, partially relevant to problem statements. • Scope of Study partially relevant to research objectives, and fairly described	• Research objectives are clear and relevant to the problem statement • Scope of Study are relevant to research objectives and sufficiently described	• Research objectives are clear and justified towards answering the problem statement • Scope of Study are relevant to research objective and clearly described, highlighting the limitation of the study	**x 1**	

FYP II – DISSERTATION

Evaluation criteria		Very Weak (1)	Weak (2)	Fair (3)	Good (4)	Very Good (5)	Multiplier	Marks
Literature Review (10)	**Critical analysis, relevancy and recentness of literature**	Irrelevant literature review	Literature review explains previous study, without connecting it to the current study. Lack in critical analysis.	Literature review explains previous study relevant to the current study, however, lack in critical analysis.	Literature review provides good discussion on previous study relevant to the current study, and sufficiently highlights the problem statement.	Literature review is comprehensive, critically analyse previous work relevant to current studies, with proper cross referencing. Clearly established the importance of the current work.	**x 1**	
	Quality of references (journals/conferences/etc.)	Insufficient references, none from published journal / conferences	Insufficient references, minimal number of references from published journal / conferences	Adequate references, some are relevant, and from published journal / conferences.	Sufficient references, almost all are relevant, and almost all are from published journal / references	Sufficient references and all are relevant, and almost all from published journal / references	**x 1**	
Research methodology (5)		Insufficient and poorly described methodology	Methodology is unsuitable to achieve the objectives	Methodology is partially relevant to achieving the objectives	Methodology is relevant to achieving objectives and fairly described and achievable within the time frame	Methodology is relevant to achieving the objectives and comprehensively described, and highly achievable within the time frame	**x 1**	

FYP II – DISSERTATION

Evaluation criteria	Very Weak (1)	Weak (2)	Fair (3)	Good (4)	Very Good (5)	Multiplier	Marks
Result and discussion (40)	• All results do not meet the research / project objectives	• Few results validate some of the project objectives • Analysis and discussion of the results are poor or inaccurate	• Few, or most results validate some of the project objectives • Analysis and discussion of the results are correct	• Results validate most of the research / project objectives • Analysis and presentation of results are correct, but does not link to theory • Significance of results mentioned but deviation from literature is not properly discussed.	• Results validate most of the research / project objectives • Analysis and presentation of results are correct, critically analysed and linked to theory • Significance of results and its deviation from literature is sufficiently discussed.	**x 8**	
Conclusion & recommendation (10)	• Insufficient conclusion and irrelevant with objective. • No future work recommended	• Insufficient conclusion and irrelevant with objective. • Insufficient future work related to the research / project recommended	• Most conclusions are relevant with objective, but some are not clearly written. • Future work recommended is related to the research and briefly outlined.	• Provide good conclusion relevant to the objective • Future work recommended is related to the research and briefly described.	• Provide excellent conclusion that is comprehensive • Future work recommended is related to the research is well described.	**x 2**	
						Total Score	**/ 85**

UNIVERSITI TEKNOLOGI PETRONAS

FYP II – DISSERTATION

PART B

Evaluation criteria		Very Weak (1)	Weak (2)	Fair (3)	Good (4)	Very Good (5)	Multiplier	Marks
ORGANISATION AND CLARITY (15)	Standard Guidelines and formatting	Report does not follow the standard guidelines and formatting	Report that has frequent major mistakes with the standard guidelines and format	Report that has frequent minor mistakes with the standard guidelines and format	Report that has some mistakes with the standard guidelines and format	Report that complies with the standard guidelines and format	**x 1**	
	English usage and writing skills English usage and writing skills (writing skills refers to clear, coherent and systematic writing of ideas)	Very poor English and writing skills.	Poor English and writing skills with many spellings and grammar mistake	Fair English and writing skills with noticeable spelling and grammar mistake	Good English and writing skills with minimal spelling or grammar mistake	Excellent English and writing skills with no spelling or grammar mistakes	**x 1**	
	Referencing and citations	Most references are missing from reference list and did not follow style guide	Many references are missing from reference list and did not follow style guide	Some references are missing from reference list and / or most did not follow style guide	Most references are cited in the reference list, which follow the style guide	All references are cited in the reference list, which follow the style guide	**x 1**	
							Total Score	**/ 15**

Comments:

_____

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……..…

Supervisor's / Examiner's Signature

Name:

Date:

UNIVERSITI TEKNOLOGI PETRONAS

FYP II – TECHNICAL PAPER SCORE SHEET

(To be completed by Supervisor and Internal Examiner)

Student's Name : _____ Student's ID: _____ Programme: _____

Project Title : _____

Criteria for Judging Quality	Unsatisfactory (<4)	Satisfactory (4 to <6.5)	Good (6.5 to <8)	Excellent (8 to 10)	Multiplier	Marks
Abstract	No or poor abstract; insufficient introduction, objective, problem statement, methodology and result	Sufficient abstract, sufficient introduction, objective, problem statement, methodology with result mentioned	Substantial abstract, substantial introduction, objective, problem statement, methodology with result mentioned	Comprehensive abstract, comprehensive introduction, objective, problem statement, methodology with result mentioned	**x 1**	
Background of study, problem statement, objective, scope of study, relevancy and feasibility	Insufficient background of study, objectives, problem statement, scope of study, relevancy and feasibility	Sufficient background of study, objectives, problem statement, scope of study, relevancy and feasibility	Substantial background of study, objectives, problem statement, scope of study, relevancy and feasibility	Comprehensive background of study, objectives, problem statement, scope of study, relevancy and feasibility	**x 1**	
Critical literature analysis, relevancy, recentness of literature	Insufficient literature review and inadequate analysis	Sufficient literature review and adequate analysis	In-depth and up-to-date literature review and has done substantial analysis	Comprehensive and up-to-date literature review and has done critical analysis	**x 1**	
Quality of references, citation and cross referencing	Insufficient quality references with inadequate citation and cross referencing	Sufficient quality references with adequate citation and cross referencing	Substantial quality references with substantial citation and cross referencing	Comprehensive and recent quality references with comprehensive citation and cross referencing	**x 1**	
Methodology	Project methodology and activities are insufficient, unachievable with inadequate methods	Project methodology and activities are sufficient, moderately achievable with adequate methods	Project methodology and activities are comprehensive, achievable with suitable methods	Project methodology and activities are comprehensive, highly achievable with extremely adequate methods	**x 1**	
Result and discussion	Major problem with presentation of results, data is	Results presented are not sufficient to meet objectives,	Results presented are sufficient and meet the	In-depth results presented and met the objectives,	**x 2**	

	poorly organized, limited discussion of theory or relevancy of results with respect to engineering knowledge	low quality data presentation, results not critically analyzed with respect to engineering knowledge	objectives, good data presentation, results substantially critically analyzed with respect to engineering knowledge	comprehensive data presentation, results are critically analyzed with respect to engineering knowledge		
Conclusions	Conclusions are not logical and completely unrelated to the objectives, limited evaluation of significance and quality of results	Conclusions are logical but are not related to the objectives, sufficient evaluation of significance and quality of results	Conclusions are logical and substantially related to the objectives, substantial evaluation of significance and quality of results	All conclusions are logical and related to the objectives, comprehensive evaluation of significance and quality of results	**x 1**	
Standard guidelines and format	Manuscript has major formatting error	Manuscript has minor formatting error	Manuscript has a few formatting error	Manuscript complies with the given format	**x 1**	
English usage and writing skill	Poor English grammar rules and writing skills	Average English grammar rules and writing skills	Good English grammar rules and writing skills	Excellent English grammar rules and writing skills	**x 1**	
TOTAL SCORE						**/100**

Comments:

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……..…

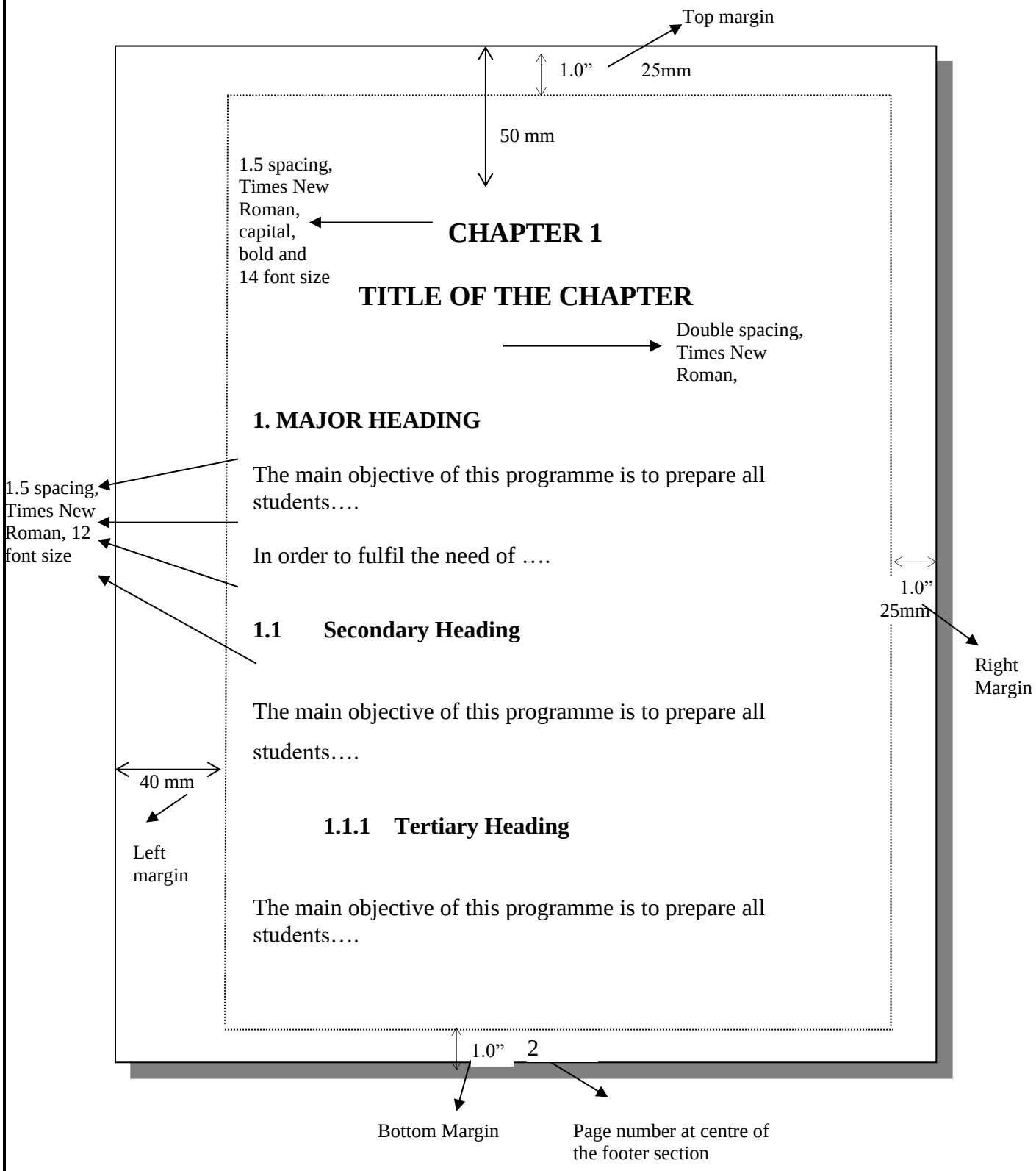
Supervisor's / Examiner's Signature

Name:

Date:

APPENDIX 4-1

SAMPLE OF PAGE SETUP



APPENDIX 4-2

SAMPLE OF TABLE AND FIGURE

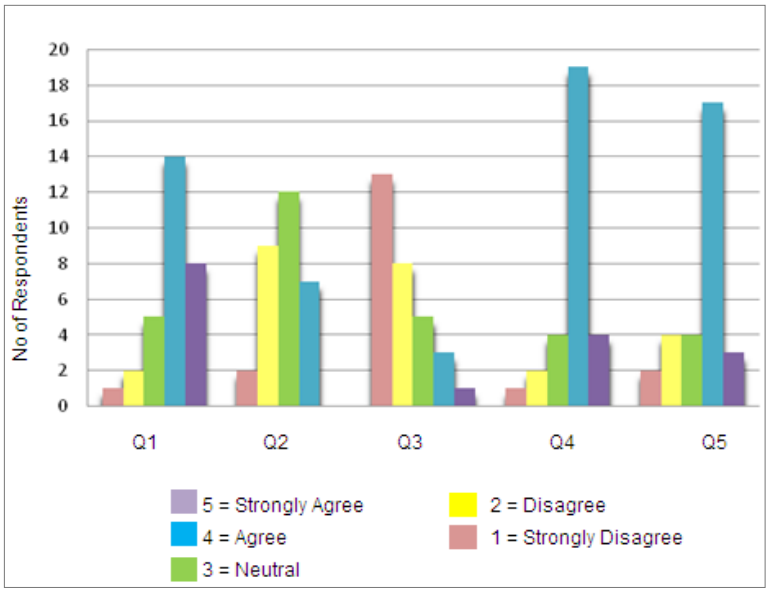


FIGURE 1. Pre-Test Survey Result

TABLE 1. Table Type Styles

Table Head	**Table Column Head**		
	Table column subhead	***Subhead***	***Subhead***
text	text	text	text

APPENDIX 5-1

SAMPLE OF DOCUMENTING SOURCES

A. The followings are the variations of APA Citations. For further reference, please refer to the internet under Documenting Sources.

1. The author's name appears as part of the introduction to the quotation or paraphrase.

   Gilster (1993) is very definite: "Simply put, the Internet is changing so rapidly, with so many new databases, services, addresses, and projects, that it can't be neatly encapsulated in any one set of commands or maxims" (p.2).

2. The author is not named in the introduction to the quotation or paraphrase.

   What is entirely clear is that the Internet "can't be neatly encapsulated in any one set of commands or maxims" (Gilster, 1993, p.2).

3. The author has several works listed in the References. If they have different dates, no special treatment is necessary; if an author has two works dated the same year, differentiate them in the text and in the References with a lower-case letter after each date (1993a,1993b).

   Gilster (1993a) points out that the Internet "can't be encapsulated in any one set of commands or maxims" (p.2).

4. Paraphrases are handled like quotations. Give the author's last name, the date, and the appropriate page numbers.

   Gilster (1993) says that the Internet changes so fast that you must come to see your experience with it as daily learning process.

5. When citing block quotations, the period is placed *before* the page parentheses. Do not place quotation marks before and after a block quotation. Indent the left margin 5 spaces and double-space. Do not indent the right margin.

According to Gilster (1993)

> There can be no complete printed directory of the Internet. Those who write about this globe-spanning network are destined to labour forever behind the technological wave. Simply put, the Internet is changing so rapidly, with so many new databases, services, addresses, and projects, that it can't be neatly encapsulated in any one set of commands or maxims. The more you use the Internet, the more you will realize that each day is itself a learning process. (p. 2)

B. The followings are the variations of IEEE Citations. For further reference, please refer to the internet under Documenting Sources.

For IEEE referencing, please refer to Appendix 5-2

APPENDIX 5-2

SAMPLE OF REFERENCE FORMATS

A. APA

The formats of references for the respective sources are as follows:

1. Journal

   Meguid, S.A. and Zhu, Z.H., 1995, "A novel finite element for treating inhomogeneous solids," *International Journal Numerical Methods Eng*ineering **38 (2):** 1579-1592

2. Book

   Glister P. 1993, *The Internet Navigator*, New York, John Wiley & Sons

3. Book in series

   Ochoa, O.O and Reddy, J.N. 1989. *Finite Element Analysis of Composite Laminates*, **(7),** New York, Pergamon Press

4. Article in book/conference proceedings

   Eskey, D and Grabe, W. 1988, "Interactive models for second language reading" in P. Carrell, J. Devine and D. Eskey (Eds) *Interactive approaches to Second Language Reading,* Cambridge; Cambridge University Press

5. Thesis

   Mohd Shariff, A. 1995, *Steam Regeneration of A Fixed Bed Adsorption System*, Ph.D. Thesis, Leeds University, United Kingdom

6. Interview

   Abu Bakar, R. Manager of Technical Services. Intel, Penang. Personal Interview. Dec. 14. 2003.

7. Technical report

   Wawrznek, P.A. and Ingraffea, A.R. 1991. *Discrete modelling of crack propagation: theoretical aspects and implementation issues in two and three dimensions*. Report Number 91-5, Cornell University, New York, USA

8. Website

   Duncan, Donna. 6 Sept 1998 <http://www.geocities.com/SoHo/Coffe/1652/>.

   Klein-Smith, Sarah. 6 Sept 1998 <http://members.aol.com/~sklein2/>

B. IEEE

The formats of references for the respective sources are as follows

Sample IEEE Documentation Style for References
(Monash University (2012, May) Institute of Electrical and Electronics Engineers (IEEE) style examples [online]. Available: http://www.lib.monash.edu.au/tutorials/citing/ieee.html)

References to sources should be numbered sequentially by order of mention in the text, with the number placed in brackets and printed on line (not as a super- or subscript) like [1]. The list of all references used in the text should appear in numerical order of mention at the end of the document. Further examples in this style can be found in the Institute of Electrical and Electronics Engineers site http://www.ieee.org/portal/site.

In-text references
Using this system, references are numbered in the order in which they are first cited in the text. If the same reference is cited later in the text, the same number is given. For example,
"The theory was first put forward in 1987 [1]"
"Scholtz [2] has argued that....."
"Several recent studies [1], [3], [4], [15], [16] have suggested that..."

Preferred	Acceptable
[1], [3], [5]	[1, 3, 5]
[1] - [5]	[1-5]

1. Books

   Elements of the citation:

   Author(s) First name or initials. Surname, or name of organisation, *Title of book followed by full stop if no edition statement, or comma if there is an edition statement*, ed., Edition (except the first). Place of publication City: Publisher, Year of Publication.

   In addition, to the above citation details, provide page numbers if you have quoted specific facts or materials e.g. pp. 28-30.

   Example:
   C. W. Lander, Power Electronics, 3rd. ed., London: McGraw-Hill, 1993.
   B. Hancock, *Advanced Ethernet/802.3 Network Management and Performance*. Boston: Digital Press, 1994, pp. 5-8.

2. Sections / chapters of books

Elements of the citation:
Author(s) First name or initials. Surname, "Title of the chapter," in *Title of the book*, ed., Edition (except the first) vol., volume if available, Ed. editor if available, Place of publication: Publisher, Year of Publication, pp. Chapter/s or First and Last pages of the article.

Example:
G. K. Knopf and A. S. Bassi, "Biological-based optical sensors and transducers," in *Opto-mechatronic Systems Handbook: Techniques and Applications*, Hyungsuck Cho, Ed. Boca Raton, FL: CRC Press, 2003, pp. 195-210.

3. Papers from conferences

Elements of the citation:
Author(s) First name or initials. Surname, "Title of paper," in *Title of the Conference*, Editor/s first name & last name if available, Ed. Place of publication: Publisher if available, Date of publication, pp. first and last pages of the paper.

Example:
A. H. Cookson and B. O. Pedersen, "Thermal measurements in a 1200kV compressed gas insulated transmission line," in *Seventh IEEE Power Engineering Society Transmission and Distribution Conference and Exposition*, 1979, pp. 163-167.

4. Journal articles

Elements of the citation:
Author(s) First name or initials. Surname, "Title of article," *Title of journal*, vol. volume, (issue number), pp. first and last pages of the article, Date of issue month if available year.

Example:
K. P. Dabke and K. M. Thomas, "Expert system guidance for library users," *Library Hi Tech*, vol. 10, (1-2), pp. 53-60, 1992.

5. Theses or dissertations

The following example is taken from the IEEE Computer Society Style Guide <http://www.computer.org/author/style/index.htm>

Elements of the citation:
Author(s) First name or initials. Surname, "Title of thesis," Type of thesis PhD dissertation or doctoral dissertation or master's thesis, Department, University, Place, State, Country, Year of Publication.

Example:
S. Birch, "Dolphin-human interaction effects: frequency mediated psychophysiological responses in biological systems," doctoral dissertation,

Dept. Electrical and Computer Systems Engineering, Monash University, Victoria, Australia, 1997.

6. Electronic sources (Electronic book)

   Elements of the citation:

   Author(s) First name or initials. Surname. (date of publication year, month day). Title. (ed. edition except the first) [Type of medium]. *volume number if needed. (issue number if needed)*. Available: site/path/file

   Example:
   A. K. Salkintzis. (2004). Mobile Internet: *enabling technologies and services*. [Online]. Available: http://www.engnetbase.com/books/1253/1631_fm.pdf

   V. Guruswami. (2004). *List decoding of error-correcting codes: winning thesis of the 2002 ACM doctoral dissertation competition*. (2nd ed.) [Online]. 3282. Available: http://portal.acm.org/3540240519.pdf
   Note: Fictitious examples

7. Online journal article

   Elements of the citation:

   Author(s) First name or initials. Surname. (year, month). Title of article. *Title of Journal*. [Type of medium]. *volume number (issue number)*, pp. pages. Available: site/path/file

   Example:
   J. S. Fulda. (2000, Mar.). The Internet as an engine of scholarship. *ACM SIGCAS Computers and Society*. [Online]. 30 (1), pp. 17-27. Available: http://doi.acm.org/10.1145/572217.572222

   J. Farrell. (2007, May). In Wikipedia we trust? *Cosmos Online* [Online]. Available: http://www.cosmosmagazine.com/node/1339

8. Electronic conference paper

   Elements of the citation:

   Author(s) First name or initials. Surname. (year, month). Title. Presented at Conference title. [Type of Medium]. Available: site/path/file

   Example:
   X. Yang. (2003, Aug.). NIRA: a new Internet routing architecture. Presented at ACM SIGCOMM FDNA 2003 Workshop. [Online]. Available: http://www.isi.edu/newarch/DOCUMENTS/yang.nira.pdf

9. Website

Elements of the citation:

Author. (year, month). Title. [Type of Medium]. Available: site/path/file

Example:
Dr Jean Armstrong. (2007, March): Brief Biography [Online]. Available: http://www.ecse.monash.edu.au/staff/jeana/aboutarmstrong.html

Reference list - Sample format:

References must be listed in the order in which they were cited (numerical order) not in alphabetical order.

[1] C. W. Lander, *Power Electronics*, 3rd. ed., London: McGraw-Hill, 1993.

[2] B. Hancock, *Advanced Ethernet/802.3 Network Management and Performance*. Boston: Digital Press, 1994, pp. 5-8.

[3] G. K. Knopf and A. S. Bassi, "Biological-based optical sensors and transducers," in *Opto-mechatronic Systems Handbook: Techniques and Applications*, Hyungsuck Cho, Ed. Boca Raton, FL: CRC Press, 2003, pp. 195-210.

[4] A. H. Cookson and B. O. Pedersen, "Thermal measurements in a 1200kV compressed gas insulated transmission line," in Seventh *IEEE Power Engineering Society Transmission and Distribution Conference and Exposition*, 1979, pp. 163-167.

[5] K. P. Dabke and K. M. Thomas, "Expert system guidance for library users," *Library Hi Tech*, vol. 10, (1-2), pp. 53-60, 1992.

[6] S. Birch, "Dolphin-human interaction effects: frequency mediated psychophysiological responses in biological systems," Ph.D. dissertation, Dept. Electrical and Computer Systems Engineering, Monash University, Victoria, Australia, 1997.

[7] A. K. Salkintzis. (2004). *Mobile Internet: enabling technologies and services*. [Online]. Available: http://www.engnetbase.com/books/1253/1631_fm.pdf

[8] V. Guruswami. (2004). *List decoding of error-correcting codes: winning thesis of the 2002 ACM doctoral dissertation competition*. (2nd ed.) [Online]. 3282. Available: http://portal.acm.org/3540240519.pdf

[9] J. S. Fulda. (2000, Mar.). The Internet as an engine of scholarship. *ACM SIGCAS Computers and Society*. [Online]. 30 (1), pp. 17-27. Available: http://doi.acm.org/10.1145/572217.572222

[10] J. Farrell. (2007, May). In Wikipedia we trust? Cosmos Online [Online]. Available: http://www.cosmosmagazine.com/node/1339

[11] X. Yang. (2003, Aug.). NIRA: a new Internet routing architecture. Presented at ACM SIGCOMM FDNA 2003 Workshop. [Online]. Available: http://www.isi.edu/newarch/DOCUMENTS/yang.nira.pdf

[12] Dr Jean Armstrong. (2007, March): Brief Biography [Online]. Available: http://www.ecse.monash.edu.au/staff/jeana/aboutarmstrong.html

APPENDIX 6-1a

Project Title

by

Student name
Student ID number

Dissertation submitted in partial fulfilment of
the requirements for the
Degree of Study (Hons)
(Programme)

FYP II Semester and Year

Universiti Teknologi PETRONAS
32610, Bandar Seri Iskandar
Perak Darul Ridzuan

APPENDIX 6-1b

SAMPLE OF TITLE PAGE

Application of Collagen as a Filter Aid in Water Treatment Process

by

Ahmad Nawab Bin Ahmad Al-Bab
123456

Dissertation submitted in partial fulfilment of
the requirements for the
Bachelor of Engineering (Hons)
(Chemical)

JANUARY 2006

Universiti Teknologi PETRONAS
32610, Bandar Seri Iskandar
Perak Darul Ridzuan

APPENDIX 6-2

SAMPLE OF FRONT HARD COVER

INTAN BT. A. RAHIM

B. ENG. (HONS) CHEMICAL ENGINEERING

JA JANUARY 2006

EFFECT OF pH AND TEMPERATURE ON THE RATE OF MINERALS LEACHED FROM MINERAL STONES

INTAN BINTI A. RAHIM

CHEMICAL ENGINEERING
UNIVERSITI TEKNOLOGI PETRONAS
JANUARY 2006

**Font : 14 Times New Roman*

APPENDIX 7-1

SAMPLE OF CERTIFICATION OF APPROVAL

CERTIFICATION OF APPROVAL

Application Of Collagen As A Filter Aid In Water Treatment Process

by

Ahmad Nawab Bin Ahmad Al-Bab
123456

A project dissertation submitted to the

Chemical Engineering Programme

Universiti Teknologi PETRONAS

in partial fulfilment of the requirement for the

BACHELOR OF ENGINEERING (Hons)

(CHEMICAL)

Approved by,

_____

(Name of Main Supervisor)

UNIVERSITI TEKNOLOGI PETRONAS

BANDAR SERI ISKANDAR, PERAK

January 2006

APPENDIX 7-2

SAMPLE OF CERTIFICATION OF ORIGINALITY

CERTIFICATION OF ORIGINALITY

This is to certify that I am responsible for the work submitted in this project, that the original work is my own except as specified in the references and acknowledgements, and that the original work contained herein have not been undertaken or done by unspecified sources or persons.

_____

AHMAD NAWAB BIN AHMAD AL-BAB

APPENDIX 8-1

SAMPLE OF ABSTRACT

The following abstract is taken from Ramos, Juan, Florentina Davalos, and Jorge Sandoval. High-brightness CMP from *Eucalyptus globulus* using a nitric acid pretreatment. TAPPI Journal 79 (12 December 1996): 169-177. Copyright TAPPI 1996.

A high-brightness, high-yield cheminmechanical pulp was obtained from *Eucalyptus globulus* using low-environmental-impact chemical reagents. The pulping chemicals were nitric acid and sodium hydroxide, and bleaching chemical was hydrogen peroxide. Chips were impregnated for 24 h in nitric acid, cooked under variable conditions, washed, impregnated with soda for 24 h, cooked again, rewashed, defibrated, refined, screened, and finally bleached under variable conditions. Under the optimal pulping conditions identified in this study, pulp strength was not especially high (tensile strength 2.04 km, tea strength 3.9 mN m^2/g), but the ease of bleaching and final pulp brightness were impressive enough (light-scattering coefficient 49.3 m^2kg, brightness 81.3% Elrepho) to warrant further research.

APPENDIX 9-1

SAMPLE OF TABLE OF CONTENTS

TABLE OF CONTENTS

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SAMPLE OF TECHNICAL PAPER

Title (The length is less than 15 words)

Writer[1], co-writer[2] (s)[3]

Affiliation [1]… [2]… [3]…

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Abstract

Writing format for this abstract is using font type, Times New Roman, size 11 points and written in a single-spacing format. Your abstract must consist summary of this article, brief of research, finding and impact of the article must include in this abstract. The uniqueness of this article also can be written in this abstract to attract the readers' interest. Citation is not allowed in the abstract. The length of this abstract is not more than 300 words.

Keywords *Maximum of 6 (six). Please do not repeat the word in the title.*

INTRODUCTION

Article journal must follow the journal house style. Article length must not be more than 5000 words, must be submitted in MS Word and formatted to A4 size. The text must be written in Times New Roman, 1.5 spacing, justified. All the citations must follow the IEEE style [1].

Every new paragraph must be flush left with one blank line between each paragraph. Articles should consist of introduction, methodology, sampling, result, and discussion. The discussion has to be succinct. Writers are advised to write the article clearly and incoherence. Data should correctly be presented. All those citations should be cited ethically.

Subheading

Organize your writing into heading and subheading. Only up to **three levels** of subheading is allowed in your writing.

All table has to be numbered, and the title has to be written above the table. The table font size is 11 points and in a single-spacing format as the sample in Table 1.

Table 1 Title of the table

Functional Group	Wavenumbers (cm^{-1})
O-H	3423.56
C-H	2925.58
C-O	1635.57
C-O	1048.38
C-OOO	608.83

All figures, charts, photos have to be in JPEG or EPS format with 300 dpi. The legend font size is 10 points and the font type is New Times Roman. The figure's source file should be written accordingly, as shown in Figure 1.

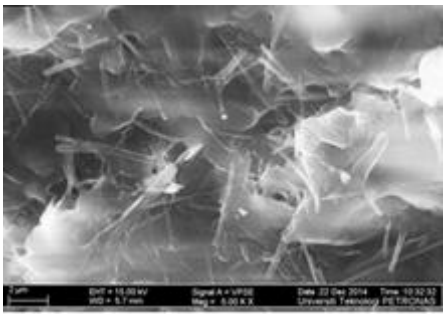


Figure 1 Figure title

Subheading

All equation has to be in Mathtype or Math Equation format. Kindly use the symbol for degree ° and not using o with superscript ^{o}.

CONCLUSION

To conclude the overall discussion of the paper. Future research may put in.

ACKNOWLEDGMENT

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REFERENCES (in IEEE style)

[1] Author Name, “Article title”, Journal/Publication title, issue, vol, no, pages, year. Doi(if any)

[2] D.W.O' Connell, C. Biekinshaw & T.F. Dwyer, "Heavy Metal adsorbent prepared from the modification of cellulose: A review”, *Biosource Technology,* 45, 2, pp. 6709-724, 2008.

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